Technical Document #731: Blockchain - Comprehensive Overview

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Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.

- Smart contracts are self-executing contracts with terms directly written into code.

- Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions.